



W3A Foundation

Investor Memorandum: Acquisition of U.S. Patent 11,803,853 and Consolida- tion of the Post-Quantum IP Portfolio

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Abstract

Lux Industries proposes to acquire the patent family anchored by U.S. Patent No. 11,803,853 B2 (“Controlling transactions on a network”) from Nahmii AS of Bergen, Norway, and to place it — together with all Lux post-quantum cryptography filings — in a single wholly-owned IP holding entity, the **W3A Foundation**. The family carries a May 2018 priority date, three granted patents (two U.S., one European), a clean chain of title, no encumbrances of record, and roughly thirteen years of remaining term. Its claims sit at the transaction-control layer that every Layer-2 network, cross-chain bridge, and threshold-custody flow passes through; Lux’s post-quantum portfolio sits at the signature layer beneath it. Held together, the two layers form a defensive moat and a licensing surface neither provides alone.

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1 Executive Summary

- **Target asset.** The “Controlling transactions on a network” family (inventor Mark Briscoombe): US 11,803,853 B2, US 12,586,073 B2, and EP 3 788 757 B1, all granted, priority 4 May 2018, expiring on or about 6 May 2039.
- **Seller.** Nahmii AS (org. nr. 991482741, Bergen, Norway; formerly Hubii AS). Active and solvent per the Norwegian business registry — this is a negotiated purchase from a going concern, not a distressed sale.
- **Status.** All three patents in force. Title of record is clean; no security interests, litigation, or post-grant challenges found. The family was extended as recently as 24 March 2026, when the U.S. continuation granted — the seller has kept it alive and current.
- **Structure.** The family is acquired by, and all Lux post-quantum patents and applications are assigned into, the W3A Foundation — a new IP holding entity wholly owned by Lux Industries, Inc. Operating entities take a perpetual intra-group license back.
- **Near-term obligation.** The first U.S. maintenance fee on the ’853 patent falls due in the window opening 31 October 2026. The acquirer must docket it at closing.

2 The Asset

2.1 Patent family

Right	Application	Granted	Status	Notes
GB 1807389.0	filed 2018-05-04	—	priority	Root priority filing
PCT/EP2019/061595	filed 2019-05-06	—	ceased	WO 2019/211494; national phase complete
US 11,803,853 B2	17/052,944	2023-10-31	in force	National-stage entry 2020-11-04
US 12,586,073 B2	18/384,566	2026-03-24	in force	Continuation of 17/052,944
EP 3 788 757 B1	19722865.3	granted	in force	Validated states to be confirmed

Table 1: The ’853 family. Both U.S. patents share the 20-year term measured from the PCT filing date: expiry on or about 6 May 2039 (subject to any patent-term adjustment and terminal disclaimers, to be confirmed in diligence). Approximately 12.8 years of term remain.

2.2 What the claims cover

The specification describes, and the claims recite, a network architecture in which transaction *origination* is separated from transaction *execution*, with execution gated on independent validation:

1. User nodes originate transactions through a server rather than directly against the chain.
2. **Validator nodes** independently verify both the *completeness* of transaction data (defeating data-withholding attacks) and its *correctness* (defeating fraud).
3. An **oracle** (second agent) is queried for the validation outcome; a **smart contract** (first agent) permits execution only on a passing result.
4. Covered transaction types include deposit, payment, settlement, withdrawal, and balance requests conducted over state channels, with remedial actions (pause, suspend, terminate, alert) on detected fraud.

Two properties give the family its weight. First, the **priority date**: May 2018 predates the deployment of essentially every modern rollup, optimistic or validity-based, every production cross-chain bridge, and every oracle-gated settlement system now in use. Second, the **grant record**: the EPO — which does not grant abstract business methods — allowed the European counterpart, which supports the technical character of the claims and is a meaningful (though not dispositive) comfort signal against U.S. §101 challenges.

3 Status and Chain of Title

This section answers the threshold question: where does the asset stand today, and who can sell it.

3.1 Ownership

- **Owner of record**: Nahmii AS, Bergen, Norway.
- **Chain of title**: inventor Mark Briscoe → Hubii AS (assignment recorded 19 November 2020) → Nahmii AS (change of name recorded 28 February 2023). The 2023 event is a renaming of the same legal entity, not a transfer — the corporate lineage in the Norwegian registry is Mapitall AS (2007) → Hubii AS (2011) → Nahmii AS (2021), one org number throughout.
- **Encumbrances**: no security interests, licenses, or liens appear in the public legal-events record. To be confirmed against the USPTO Assignment database and EPO register at diligence.

3.2 Seller condition

Nahmii AS is registered active: not bankrupt, not under voluntary or compulsory liquidation. The company is a Series-A Ethereum Layer-2 infrastructure firm (roughly \$8M raised) and was selected by Norges Bank for its experimental CBDC sandbox. Two consequences for deal posture:

- There is no distress discount to harvest. Price discovery will be a negotiation with a functioning management team that recently spent money extending the family (the March 2026 continuation grant).
- The seller will almost certainly require a **license back** for its own L2 operations. This is acceptable: the strategic value to Lux is defensive coverage and portfolio anchoring, not excluding Nahmii.

3.3 Maintenance posture

- **US '853**: first maintenance fee (3.5-year) window opens 31 October 2026; due 30 April 2027, surcharge period to 31 October 2027. *This is the first post-closing docket item.*
- **US '073**: first maintenance fee due 2029.
- **EP '757**: annual annuities in each validated state; current status per state to be confirmed.
- **Litigation / post-grant**: no district-court cases, ITC actions, IPRs, or PGRs located. Formal docket and PTAB searches are a diligence item.

4 Strategic Rationale

4.1 Two layers, one moat

Lux’s post-quantum work protects the *signature layer*: threshold ML-DSA (Pulsar), R-LWE threshold signing (Corona), threshold SLH-DSA (Magnetar), the Quasar post-quantum consensus family, on-chain post-quantum verification precompiles (P3Q), and GPU-native verification. The ’853 family protects the *control layer* above it: validator-gated, oracle-verified transaction execution — the architectural pattern through which every L2 sequencer, bridge attestation, and MPC-custody withdrawal actually flows.

A post-quantum signature portfolio alone invites the response “design around the scheme.” A transaction-control patent alone invites “design around the oracle.” Held together under one entity, they cover both what signs and what gates — and the 2018 priority date of ’853 backstops the whole position in a way no new filing can reproduce.

4.2 Uses

1. **Defensive.** Freedom-to-operate insurance and counter-assertion mass for Lux Network, its bridges, and MPC custody. Foundation ownership insulates the operating companies from direct exposure of the crown-jewel IP.
2. **Licensing surface.** The foundation can run a standard program: royalty-free defensive pledges to the open-source ecosystem, commercial terms for proprietary L2/bridge operators. The EPO grant gives the program credibility in Europe.
3. **Portfolio anchor.** A granted 2018-priority family materially raises the floor value of the consolidated portfolio versus a collection of young, application-stage PQ filings.

5 Holding Structure: W3A Foundation

- **Entity.** W3A Foundation, wholly owned by Lux Industries, Inc. To make “wholly owned” legally exact, the recommended form is a foundation company with Lux Industries as sole member (or, equivalently, a conventional IP-holdco subsidiary bearing the Foundation name). Formation has not yet occurred; no public registration exists as of this memo. Entity formation is a condition precedent to closing.
- **Consolidation.** At or promptly after closing: (i) the ’853 family assigns from Nahmii AS to W3A; (ii) all Lux post-quantum patents, applications, and invention disclosures assign from their current holders into W3A; (iii) all assignments are recorded at the USPTO and EPO.
- **Operating licenses.** W3A grants a perpetual, irrevocable, royalty-bearing (or capital-contribution-offset) license to Lux operating entities; external licensing runs through the foundation exclusively. One entity, one licensing surface, one place to diligence — and one place for a future acquirer or investor to value.

6 Transaction and Valuation Frame

No price has been discussed with the seller; the figures below are a pre-diligence working frame, not an offer basis.

- **Comparables.** Granted blockchain-infrastructure families with U.S. + EP coverage and 10+ years of term have historically cleared in the mid-six to low-seven figure range (USD) in negotiated sales; single-patent lots without EP counterparts trade materially lower. The

March 2026 continuation grant and the 2018 priority date argue for the upper half of that band; the absence of any enforcement history argues for the lower half.

- **Structure.** Cash at closing for the assignment, plus a royalty-free, non-transferable grant-back to Nahmii AS limited to its own operated networks. Avoid earn-outs tied to enforcement — they entangle the seller in future assertion decisions.
- **Cost basis after closing.** Maintenance and annuities across the family run low four figures annually in the near term (rising at the 7.5- and 11.5-year U.S. marks); diligence, recordation, and transfer costs are one-time and modest relative to purchase price.

7 Risk Factors

1. **§101 (Alice) exposure.** U.S. claims directed to transaction control carry inherent abstract-idea risk. Mitigants: claims recite a concrete network architecture (validator nodes, oracle, execution gating), and the EPO granted the counterpart. Risk is real but priced into the valuation band.
2. **Prior art.** The 2016–2018 state-channel and plasma literature (Lightning, Raiden, Plasma) is the natural invalidity hunting ground. An invalidity-grade prior-art search is required *before* an offer is priced, not after.
3. **Family may be closed.** If Nahmii filed no further continuation before the '073 grant on 24 March 2026, no new U.S. claims can be pursued (a broadening reissue of '073 remains available until 24 March 2028). Whether a continuation is pending is a first-order diligence question — an open family is worth meaningfully more.
4. **Design-arounds.** The claims require the validator/oracle interplay as recited; architectures that collapse validation into the execution contract may avoid them. Value accordingly rests on the portfolio position and priority date, not on any single unavoidable claim.
5. **EP validation gaps.** The European patent may be validated (or still in force) in fewer states than assumed; annuity lapses shrink the European position.
6. **Enforcement reality.** Asserting against decentralized networks is structurally hard (who is the infringer?). The thesis here is defensive depth and licensing, not assertion revenue — investors should not underwrite litigation upside.

8 Diligence Plan and Next Steps

1. **Title and encumbrance:** full USPTO Assignment Search and EPO register pull; confirm no liens, exclusive licenses, or springing interests; obtain the Briscoe inventor-assignment instruments.
2. **Family completeness:** USPTO Patent Center review for pending continuations, terminal disclaimers, and patent-term adjustment on both U.S. patents; EP register for validated states and annuity status.
3. **Validity:** commissioned invalidity-grade prior-art search over 2015–2018 state-channel, sidechain, and oracle literature; §101 opinion from U.S. counsel.
4. **Adversarial record:** PTAB, district-court, and ITC docket sweeps for any challenge or assertion history.

5. **Seller engagement:** approach Nahmii AS through Norwegian counsel; establish whether IP monetization is on their agenda and whether the grant-back structure lands.
6. **Entity formation:** form W3A Foundation with Lux Industries, Inc. as sole member; open the docketing calendar with the 31 October 2026 '853 maintenance window as the first entry.
7. **Consolidation:** execute and record the intra-group assignments of the Lux post-quantum portfolio into W3A concurrently with closing.

Sources: USPTO / Google Patents records for US 11,803,853 B2, US 12,586,073 B2, WO 2019/211494 A1, and EP 3 788 757 B1; Norwegian Business Registry (Brønnøysundregistrene) entity 991482741. All patent-office facts verified July 2026; diligence items flagged above remain open.